

Note, for example, that every specific convergent sequence has only one limit. It means that the words "the limit of a given sequence" are self-explanatory. As for a function defined over an interval, one can speak of an infinite number of "limits" because the limit of function is found for each specific point $x=a$ or, as we say, for x tending to a . Thus the phrase "the limit of a given function" is meaningless because "the limit of a given function must be considered ~~for~~ ^{only at} each given point a ". Besides, this point a should either belong to the domain of the function or coincide with one of the ends of the domain.

Reader: In this case the definition of the limit of function should be very different from that of the limit of sequence.

Author: Certainly, there is a difference. Note, first of all, that we analyze a function $y=f(x)$, which is defined over a segment, and a point a in this segment (which may coincide with one of its ends when the function is defined over an open or half-open interval).

Reader: Do you mean to say that at the point $x=a$ the function $f(x)$ may not be defined at all?

Author: That is quite correct. Now let us formulate the definition of the limit of function.

Definition: A number b is said to be the limit of a function $f(x)$ at x tending to a (the limit at point a) if for any positive value of ε there is a positive value of δ such that for all x satisfying the conditions x belongs to the domain of the function; $x \neq a$ and $|x-a| < \delta$ (1) we have $|f(x)-b| < \varepsilon$ (2).

The standard notation is $\lim_{x \rightarrow a} f(x) = b$. Imp. Remember

Reader: The definition of the limit of function is noticeably longer and more complicated than that of the limit of sequence.

Author: Note, first of all, that according to (1), point x should belong to the interval $[a-\delta, a+\delta]$. Point $x=a$ should be eliminated from this interval. The interval $[a-\delta, a+\delta]$ without point $x=a$ is called a punctured δ -neighbourhood of point a .